DOMAIN 8
TRAINING AND EDUCATION (QUESTIONS & ANSWERS)

2089. The steps to successful training program are which of the following ?

1) Analysis, design, implementation, evaluation
2) Site survey, course development, classroom setup, testing
3) Requirements definition, gap assessment, content development, implementation
4) Proposal development, pre-testing, content development, post-testing

Ans.1 The steps to accomplish a successful training action are: Analysis, Design/development, Implementation, Evaluation/improvement.

Mager, R.F. (1997). Preparing Instructional Objectives, (3rd ed.). CEP Press. 2.

2090. Which of the following is an example of a program sponsored by an Employee Assistance Program?

1) Smoking cessation
2) Occupational counseling
3) Substance abuse counseling
4) All of the above

Ans.4 Employee Assistance Programs (EAPs) have gained popularity as a management system to retain good employees by assisting them with their personal problems that cause occupational problems. For example, in many cases an employee with an alcohol or other substance abuse problem can enroll in the appropriate EAP program for treatment without the risk of termination provided that the employee meets all of the requirements of the program.

2091As the facility safety professional, you have become acquainted with two line employees that want to learn more about occupational safety in their own work environments beyond what is typically provided to them in their standard training courses. Which of the following is the best way to encourage their continued interest in safety?

1) Work with the employees and their supervisors to develop mutually agreeable written individual development plans outlining new training goals and safety-related responsibilities.
2) Work with their supervisors to send them to advanced occupational safety training.
3) Work with the employees and their supervisors to permit them to accompany you for a week.
4) Work with their supervisors to assign these employees as safety coordinators in their respective operations.

Ans.1 Individual development plans (IDPs) are agreements between employees and their managers to help increase the employees' satisfaction with their jobs. IDPs should include measurable performance goals for attending training, demonstrating new skills, and achieving more diverse job responsibilities.

2092. In a certain modern high-hazard laboratory, employee identification badges are color-coded. Five different colors are applied in groups of three. Each group of three colors represents one of the possible occupational hazards for which a worker has been properly trained. How many types of occupational safety training can be represented?

1) 10
2) 60
3) 120
4) 240

Ans.2 When the order of selection is significant then it is a permutation.

Permutations = N! / (N - X)!

(where N is the total number of colors and X is the number in a single grouping)

Permutations = 5! / (5-3)! = 120 / 2 = 60

There are 60 possible types of occupational safety training that can be displayed on an employee badge.

2093. You passed the CSP Examination with a score of 192. The cumulative statistics for this examination are as follows: The mean score is 180 and the standard deviation is 32. Assuming a standard distribution, how many safety practitioners taking the exam scored higher than you scored?

1) 35.4%
2) 37.5%
3) 40.2%
4) 64.6%

Ans.1. $z = (X - \bar{x}) / s = (192 - 180) / 32 = 0.375$

With a z of 0.375, refer to the Table of the Normal Distribution and determine the area under the curve associated with a z of 0.375.

To do this, select 0.3 under the "z" column. Go over to the "7" column along the row of z = 0.3 (representing a z of 0.37). The area for z = 0.37 is 0.1443. The area for z = 0.38 is 0.1480. Interpolate these two numbers (take the average). This is the area under the curve for z = 0.375.

Area = (0.1443 + 0.1480) / 2 = 0.14615

Since we are interested in ALL folks who have scored higher than you, we have to add the area under the curve less than the mean to the area we just calculated that falls above the mean. This means that we have to add 0.50000 to the area we calculated above.

Total Area = 0.50000 + 0.14615 = 0.64615

The percentage of folks who scored higher than you scored = 1 - Total Area
= 1 - 0.64615
= 0.35385
= 35.4% scored higher

2094. Which of the following is considered to be the least effective method for group teaching?

1) Demonstration
2) Role play
3) Lecture
4) Discussion

Ans.3. Demonstration is especially effective when a language problem exists. Role playing allows participants to try different behaviors without a great deal of risk, providing immediate feedback. Discussion allows students to actively participate and lends itself toward greater behavioral modification.

2095. Training is recognized as a powerful influence and motivator in safety. When developing a safety training program, the first consideration should be to:

1) Acquire a training staff
2) Determine the scope and content of the training program
3) Define the training objectives
4) Obtain training materials

Ans.3. Objectives should always be spelled out before the content, methods, and staffing can be selected. These objectives should be reduced to terms of behavior; in other words, what is it that the training participants should be able to accomplish when their training has been completed.

2096. The best fire extinguisher training course will have what components?

1) Dynamic classroom instruction using extensive graphics and pictures
2) Classroom instruction with instructor-performed demonstrations of actual fire extinguishers
3) Classroom instruction with instructor-performed demonstrations and instructor-supervised student use of actual fire extinguishers on controlled fires
4) Instructor-performed demonstrations along with instructor-supervised student use of actual fire extinguishers on controlled fires

Ans.3. The best fire extinguisher training course will couple classroom instruction with actual demonstrations leading to student use of fire extinguishers on real fires. Adults learn effectively with practical exercises.

2097. When confronted with a problem with which you have little practical experience, which of the following problem-solving schemes is likely to produce the best resolution?

1) Establishing a committee of subject matter experts (at least 15) to discuss ways to resolve the problem and come to a consensus
2) Establishing a small group of subject matter experts (at least 3) to discuss ways to resolve the problem and come to a consensus
3) Performing your own thorough research and resolving the problem yourself
4) Delegating the problem to another person with experience and training similar to your own for resolution

Ans.2. Though all of these could ultimately lead to the best resolution, the dynamics of a small group of subject matter experts typically offer the best foundation for solving problems.

2098. After any initial ice-breaking/rapport-building sessions, all training courses should start with:

1) The course's objectives
2) The course's break schedules
3) The course's daily adjournment expectations
4) A description of the course's sponsor

Ans.1. Though all of these would be appropriate in some training courses, not all training courses will have them all. On the other hand, all training courses should have course objectives very clearly defined by the instructor at the beginning of the course.

2099. Evaluate the following course objectives that are on the instructor's first slide:

- To provide the student with the tools necessary to understand and feel comfortable with working in an environment containing hazardous chemicals such as acetone, methanol, methyl ethyl ketone, muriatic acid, lye, and iso-propanol
- To provide the student with the confidence to effectively understand the contents of the company's material safety data sheets and the contents of the company's inventory of hazardous chemicals

- To provide the student with the knowledge necessary to understand the types of hazards posed by the various chemicals listed above and how the student can protect himself or herself from exposure to these hazards using personal protective equipment
- To provide the student with information on OSHA's hazard communication standard and the company's hazard communication program as required by 29 CFR 1910.1200.

1) The objectives are adequate.
2) The objectives are not adequate because they do not adequately define all of the necessary hazard communication training objectives.
3) The objectives are not adequate because they are too detailed and wordy.
4) The objectives are not adequate because they are insufficiently detailed for the student.

Ans.3. Objectives at the beginning of the course should be as short as possible to help the students mentally navigate through the course. Very few details should be given in objectives - that's what the rest of the training should provide.

2100. Students should be given the opportunity to demonstrate competence in a new skill in the presence of an instructor. Is it a good idea for the instructor to "test" students performing these practical exercises against objective criteria?

1) Yes
2) No, students have sufficient pressure performing the skill without the added stress of being tested.
3) No, an instructor cannot fairly gauge a student's practical performance against objective criteria.
4) No, students will be unfairly competing with one another in a testing environment.

Ans.1. Without question, an instructor should evaluate a student's practical performance against objective criteria for successfully performing the skill.

2101. A multiple-choice quiz that requires the student to demonstrate an understanding of the knowledge from a course is an example of what kind of test?

1) Subjective
2) Objective
3) Random
4) Sequential

Ans.2. Multiple-choice quizzes are typically objective. The difference between an objective and a subjective multiple-choice test is how it would be graded by different people with the same knowledge as presented by the instructor. A student's score on an objective test will be the same no matter who grades it because the facts are absolute and not subject to an instructor's interpretation.

2102. An essay test that requires the students to demonstrate an understanding of the two objectives from a course is an example of what kind of test?

1) Subjective
2) Objective
3) Random
4) Sequential

Ans.2. Even though this is an essay test, it is still objective because the essay question is limited to an explanation of the objectives from the course. These objectives are based on the facts that the instructor presented and do not require much in the way of subjective interpretation. There could be some subjectiveness in the grading, however, this test would be mostly objective.

2103. An essay test that requires the students to interpret the meaning of a philosophy discussed during the class is an example of what kind of test?

1) Subjective
2) Objective
3) Random
4) Sequential

Ans.1. This is an example of a subjective test. The interpretation of a philosophy is subjective to both the student and the instructor. There will likely be very broad differences in grading based on the specific experiences of the graders.

2104. Modern technology exists for presenting visual images to students. A popular example of more modern visual imaging devices is the:

1) Opaque projector
2) Computer video projector
3) 35 mm slide projector
4) Overhead projector

Ans.2. Computer video projectors have gained popularity due to their light weight, affordable prices, and their ability to present colorful images without the need to physically print or produce an image from film or on transparencies.

2105. With the proliferation and advancement of computer technology, instructors have access to sophisticated software tools for generating visual and audio aids (multimedia presentation software). Two commercial examples of this software include:

1) Microsoft PowerPoint and Lotus 1-2-3
2) Microsoft PowerPoint and Corel Presentations
3) Microsoft Excel and Microsoft Windows
4) Corel Presentations and Microsoft Outlook

Ans.2. Microsoft PowerPoint and Corel Presentations are two of the most popular multimedia presentation software packages. When connected to a computer video projector, these software packages assist instructors with their presentations by providing colorful graphics and pictures along with video and audio clips and animations.

2106. When conducting presentations, there is usually confusion whether it should be done primarily using auditory or visual signals. Which one of the following is NOT true:

1) When the message is simple - use auditory signals
2) When the message is short - use auditory signals
3) When the message will be referred to later - use visual signals
4) When the message deals with events in time - use visual signals

Ans.4. When the message deals with events in time, one should primarily use auditory signals. When the message deals with events in space, visual signals should primarily be used.

2107. Besides expertise with the instructional material, an effective instructor must consider:

1) The time of day in which the class will be held
2) The layout of the room in which the class will be held
3) The names of the students in the class
4) The background and informational needs of the students

Ans.4. Though all of the above can help, an effective instructor must consider the background and the informational needs of the students to effectively present the material.

Background includes educational background, experience, language skills (for speakers of languages not native to the instructor), and positions within the company. Informational needs are also critical. A class full of high-level managers will need far less detail on specific operational issues but may need much more detail on legal and regulatory issues and other business management information.

2108. In most cases, safety-related training courses for managers will be _____ than a similar course for line/operations employees.

1) Shorter
2) Longer
3) More detailed
4) None of the above

Ans.1. In most cases, the mana ger's awareness component of a safety-related class will be shorter. In these components, managers are interested in the course's objectives and what they need to do (as managers) to ensure that the course's objectives are implemented by their employees.

2109. What is the best technique to explain a process flow or other sequential operation to employees with a low level of formal education?

1) Classroom lecture
2) One-on-one explanation
3) Visually, with few words
4) None of the above

Ans.3. Where possible, visual cues and explanations should be provided to employees who have a low level of formal education (i.e., employees who are functionally illiterate). These can be explained; however, the visual explanation should be sufficient.

2110. CBT is:

1) A system for assessing employee aptitude by computer
2) A system for delivering computer-based training to employees
3) A method for characterizing employee behavior traits
4) A strategy for calculating the employee's basic technical skills

Ans.2. Computer-based training is now an accepted method for delivering training to employees. Site-specific information can easily be adapted into more generic training programs, and employees' understanding of the material can be evaluated easily.

2111. In a training course, you give a quiz at the end of the course to assess the students' understanding of the course content. Several of your students score 68% while the rest scored between 95% and 100%. What should you do next?

1) Require them to retake the course and retake the quiz.
2) Require them to retake the sections of the course containing the material that they missed and require them to retake the relevant section of the quiz.
3) Research the questions they missed and discuss the quiz questions and course content with the students before determining any additional activities.
4) Pass the students.

Ans.3. In many cases, course quizzes have questions that can be misinterpreted by some students. Also note that a 70% is not necessarily an adequate raw score. Some quizzes may require a 100% raw score to pass while others may require a 60% raw score. Quizzes should be considered holistically with respect to the objectives of the course.

2112. Safety professionals frequently require feedback to understand the effectiveness of various safety-related management systems. Which of the following is a typical method for obtaining feedback on management systems?

1) Responses to written surveys
2) Personal discussions
3) Discussion groups
4) All of the above

Ans.4 There are several methods for obtaining feedback. Sometimes, the feedback is obtained formally, sometimes it's obtained informally.

2113. On a quiz you developed and delivered to assess the students' understanding of the course material, you noticed that 15 out of your 30 students missed the same question. Your first action should be to:

1) Rewrite this question as it is obviously poorly written.
2) Abandon this question from this quiz and rescore the students' quizzes.
3) Discuss this question with a sample of students from the class (including those who missed it and those who did not).
4) Ignore the statistical anomaly.

Ans.3 Though you could perform a formal statistical treatment on this quiz question to officially determine that it is anomalous, your gut tells you that when half the class misses the same question, there is likely a system error in either the test or the instruction. Before you rewrite the quiz question, you have to know more about the problem. Is it the wording in the question? Did you cover the material in class? The answers to these questions will likely come from a discussion with a sample of students from the class.

2114. What type of formal statistical treatment would you apply to an item on a training quiz to determine whether the item was statistically missed more frequently than would be statistically expected?

1) Σ-test
2) σ-test
3) β-test
4) χ^2-test

Ans.4 A χ^2-test would be used to determine the statistical significance of a test problem that was observed as missed more than was expected.

2115. Of the following, which would be considered the minimum number of exam items from which a viable statistical sampling could be obtained?

1) 2
2) 3
3) 5
4) 20

Ans.4 Most statisticians would agree that a sample of between 15 and 20 items is the minimum number of points from which viable statistical sampling can be performed.

2116. Your test instructions are as follows:

"This test has 25 multiple-choice questions. Each question has exactly one correct answer selection and four incorrect answer selections. Choose the correct answer to each question from the five answer selections. You will receive 2 points if you select the correct answer for the question. You will receive 0 points if you select no answer (i.e., leave the question blank), and you will have 1 point deducted for each wrong answer you select."

One of your students correctly answered 22 questions and incorrectly answered 2 while leaving one blank. What is this student's percentage score?

1) 84%
2) 86%
3) 88%
4) 97%

Ans.1 The total points that are available are 50 points (25 questions * 2 points per question). This student earned 44 points from correct answers (22 questions * 2 points per questions) but lost 2 points from incorrect answers (2 wrong questions * 1 point per wrong question). Therefore, his raw score was 42 points (44 points - 2 points = 42 points). The percentage score is therefore 84% (42 points/50 points = 0.84 = 84%)

2117. Which one of the following statements is NOT true concerning conducting presentations?

1) A moving person should use auditory signals
2) A stationary person should use visual signals
3) If the message calls for immediate action, auditory signals should be used
4) When the receiving location is noisy, auditory signals should be used.

Ans.4 When the receiving location is noisy, anything BUT auditory signals should be used.

2118. Which of the following technologies supports duplex voice communication?

1) Personal pagers
2) Trunked radio systems
3) Conventional radio systems
4) Cellular phones

Ans.4 Duplex communication is supported when all parties can hear all other parties and speak to all other parties at the same time. A telephone is an example of a device that supports duplex communication. Conventional and trunked radio systems are typically simplex (i.e., only one person can talk at a time).

2119. Software "suites" have become popular both at home and at workplaces to help automate record keeping chores. What main types of software are typically found in a standard business software suite marketed to homes and businesses?

1) Word processor, spreadsheet, database
2) Scanner software, printer software
3) Financial accounting software, scanner software
4) Modem software, operating system, human resources software

Ans.1 Frequently, more than just word processing, spreadsheet, and database software are included in software suites. Multimedia presentation software, e-mail, Internet, and other software are frequently included as well.

2120. By what mechanisms can existing data be input into a computerized data logging system?

1) Manually, via a keyboard
2) Electronically, via a scanner and optical character recognition software
3) Electronically, via an interface with another computer on which the data resides
4) All of the above

Ans.4 There are hundreds of ways to obtain data and make it reside on a specific computer. Using local and wide area network technology, as well as the Internet, computers on the other side of the world can provide data to a specific computer. Raw data can be processed by a computer through sensors and sent to a central computer for storage and distribution. Images and sound can even be captured, categorized, and stored.

2121. A colleague e-mails you the following line of text:

http://www.osha-slc.gov/OshStd_toc/OSHA_Std_toc_19 10.html

This is an example of a(n):

1) E-mail address
2) Internet address via file transfer protocol
3) Internet address via hypertext transfer protocol
4) Hexadecimal address

Ans.3 A hypertext transfer protocol string such as this one is commonly called a "web address" or "website." There are several protocols for transferring information through the Internet. Each has advantages and disadvantages. In this particular case, the Internet address points to the table of contents of OSHA's General Industry Safety and Health Standards (those standards associated with 29 CFR 1910).

2122. On your computer, you have a digital image of a hazardous condition you noticed last week. You want to immediately send this image to your peer at another facility. Both of you have e-mail. You attach the computer file containing the image to the body of your e-mail. In which of the following image formats would you likely send the image?

1) JPEG or GIF
2) FTP or VBS
3) EXE or COM
4) DLL or FOT

Ans.1 There are other ways to attach the image. For example, you could insert the image into a word processor document and send the word processor document instead of just the image. You could also send the image in several proprietary formats if you knew that your peer's computer had the capability to read these formats. JPEG and GIF formats are nearly universal, and these formats should be used for images if you are unsure of the capabilities of the receiver's computer.

2123. When teaching a class, body language is important. Regarding attire, which of the following is most appropriate?

1) Dress the same way as the majority of the students so that you "fit-in" with the students. If your students are dressed very casually, then you should be dressed very casually.
2) Dress "one notch better" than your students. If your students are dressed very causally, then you should dress business casual - if your students are dressed business casual, then you should wear a tie (man) or a suit (woman), etc.
3) Dress formally for all training.
4) Dress is completely irrelevant when leading a training class.

Ans.2 Most experts will say that the instructor should be dressed better than the majority of his or her students, but not to the point where the instructor looks out of place. Dressing slightly better than your students may influence your students to consciously and unconsciously respect you as an expert and also allow you to stand out so that you are easily identified during breaks.

2124. Which of the following is true regarding communication?

1) Both the sender's and the receiver's body language is very important in one-on-one communication.
2) Visual communication is inappropriate in groups of ten or more.
3) Visual communication is inappropriate in groups of twenty or more.
4) Body language is only important in one-on-one communication.

Ans.1 Some experts say that body language is as important as the actual content of the verbal communication during communication - regardless of whether the communication is one-on-one or to an audience. Visual communication is appropriate for any number of people as is verbal communication.

2125. Didactic model is the teaching model that:

1) presents a straight forward direction of learning, instructing the participants on what to do and how to do it.
2) provides for the accomplishment of quantitative testing results.
3) is non-judgmental.
4) presents open ended discussion topics based on the instructor's experiences, allowing a free flow of thought by the participants.

Ans.1 Didactic instruction refers to the teaching of information, often morally or philosophically based.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 25.

2126. Increasing participation of the participants is an ongoing objective of good trainers. What do all trainers always have control of to help increase participation?

1) The physical training environment
2) Interpretive skills
3) Participant expectations
4) Group interaction

Ans.2. A trainer only has control over their own actions: communication skills, interpretive skills, personal traits, interpersonal style, learning techniques, attending skills and process factors.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 151.

2127. During the lecture part of a course, the trainer works to balance eye contact with all the participants and to move around the room in a manner that encourages participation by the whole class. This would be an example of which of the following trainer controlled actions or traits to increase participation?

1) Process factors
2) Attending skills
3) Learning techniques
4) Interpersonal style

Ans.2 Attending skills of training include: providing consistent eye contact, balanced eye contact, demonstrate attentive moves, providing affirmative non-verbals and staying engaged.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 153.

2128. In an effort to build participation, a trainer starts by asking simple questions for a show of hands, building to simple short answer questions and then to discussion questions. The trainer also works to encourage participation from quiet participants. This would be an example of which of the following trainer controlled actions or traits to increase participation?

1) Process factors
2) Attending skills
3) Learning techniques
4) Interpersonal style

Ans.1 Process factors employed by trainers include: balancing participation of the participants, maintaining flow of interactions, building participation, and creating an inclusive environment.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 154.

2129. Cognitive psychology research has identified three factors influencing how well and how much we learn. Which one do we as individuals have the least capability of changing?

1) Childhood education
2) Motivation
3) Prior knowledge
4) Ability

Ans.4 Three factors that have been identified through cognitive psychology research that influence how well and how much we learn are ability, prior knowledge and motivation. Of these, learning ability is the one which is most dependent on genetic makeup.

Stolovitch, H.D., & Keeps, E.J. (2011). Telling Ain't Training, (2nd ed.). ASTD Press. 40.

2130. The determination of whether there is a problem worth solving and if training or instruction is the proper solution is part of which phase of the instructional preparation?

1) Evaluation
2) Pre-proposal development
3) Student pre-testing
4) Analysis

Ans.4 During the analysis phase of instructional preparation, the first step is to determine if there is a problem that needs to be addressed, then to make sure that training is part of the way to correct the problem. Part of analysis should also assess what type of training should be used.

Mager, R.F. (1997). Preparing Instructional Objectives, (3rd ed.). CEP Press. 2.

2131. The first step to developing training or instruction is to:

1) develop the training objectives.
2) validate the training methodology.
3) perform and initial student survey.
4) perform an analysis.

Ans.4 The first step of developing training is to complete the analysis to determine what problem exists, if training is part of the solution and what kind of instruction is appropriate. Once the analysis is complete, training objectives can be developed that can be used to guide the development of the training.

Mager, R.F. (1997). Preparing Instructional Objectives, (3rd ed.). CEP Press. 2.

2132. John has been hired by an automotive detail business to perform a job task analysis 2 weeks after the employees have been trained in the corporation's new processes for completing auto detailing. Which training evaluation level is John involved in?

1) Results
2) Learning
3) Reaction
4) Behavior

Ans.4 Behavior - measures whether the skills and knowledge learned are being implemented.

Don Kirkpatrick's four levels of training evaluation are reaction, learning, behavior, and results:

Reaction - measures participant satisfaction with the training.

Learning - measure the extent to which learning has occurred.

Behavior - measures whether the skills and knowledge learned are being implemented.

Results - measures the business impact of the implemented skills and knowledge.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 255.

2133. A manager knows training is necessary when:

1) employees are asking for training.
2) the employees need to know something that they don't already know.
3) a survey has been conducted indicating the employees are not completing tasks properly.
4) at the conclusion of an audit, the auditor has indicated training is the proper corrective action.

Ans.2 Basically, training is only effective when "something" is not known by people who have a need to know that "something". If there is no need to know the information or the skill, the training is not adding benefit.

Mager, Dr. R.F. (1999). What Every Manager Should Know about Training, (2nd ed.). CEP Press. 12.

2134. Which instruction type would be best suited to teach a new set of employees the steps to accomplish the task associated with their new position, position A8, on the assembly line?

1) Education
2) Classroom
3) Training
4) Instruction

Ans.3 Training - the purpose of training is to create a change that is consistent and reproducible.

Instruction - the purpose is to provide the learner the ability to apply what is learned in more general terms, to adapt the learning to different situations.

Education - education builds general models and value systems, occurring over a longer term, assembling a variety of experiences, learning and events.

Stolovitch, H.D., & Keeps, E.J. (2011). Telling Ain't Training, (2nd ed.). ASTD Press. 9.

2135. A new group of laborers have been hired for the day shift at the chemical plant. They are required to learn the safety rules and responsibilities related to working at the plant. Which term best describes the type or level of learning necessary?

1) Education
2) Classroom
3) Training
4) Instruction

Ans.4 Training - the purpose of training is to create a change that is consistent and reproducible.

Instruction - the purpose is to provide the learner the ability to apply what is learned in more general terms, to adapt the learning to different situations.

Education - education builds general models and value systems, occurring over a longer term, assembling a variety of experiences, learning and events.

Stolovitch, H.D., & Keeps, E.J. (2011). Telling Ain't Training, (2nd ed.). ASTD Press. 10.

2136. Numerous organizations exist creating standards and guides for everything in industry. One such organization coordinates and publishes the American Nation Standards. This organization is:

1) ANSO
2) AANS
3) USANS
4) ANSI

Ans.4 The American National Standards Institute, Inc. provides oversight and verification of the process for the development of consensus standards published as American National Standards.

American Society of Safety Engineers (2009). Criteria for Acceptable Practices in Safety, Health and Environmental Training, ANSI/ASSE Z490.1, (1st ed.). American National Standards Institute. 2.

2137. The ASSE serves as secretariat to the ANSI accredited committee that developed the American National Standard: Criteria for Accepted Practices in Safety, Health, and Environmental Training. The number of this standard is:

1) ANSI Z490.1
2) ANSI Z88
3) ANSI Z97
4) ANSI A117.1

Ans.1 ANSI Z490.1, Criteria for Accepted Practices in Safety, Health, and Environmental Training.

American Society of Safety Engineers (2009). Criteria for Acceptable Practices in Safety, Health and Environmental Training, ANSI/ASSE Z490.1, (1st ed.). American National Standards Institute. 3.

2138. Numerous standards setting bodies exist. Some are based in individual countries and some are multinational. Which of the following is a multinational standard setting organization?

1) ASSE
2) ISO
3) ANSI
4) BSI

Ans.2 ISO, the International Organization for Standardization is specifically directed at developing and publishing standards for international application.

ASSE, the American Association of Safety Engineers is a United States based organization that promotes the safety and health profession.

ANSI, the American National Standards Institute is a United States based standards development organization.

BSI, British Standards Institution is the United Kingdom's national standards body.

2139. In ANSI Z490.1, Criteria for Accepted Practices in Safety, Health, and Environmental Training, when 'should' is used in the standard it indicates a statement that is:

1) permissive.
2) mandatory.
3) a suggested requirement.
4) a recommendation.

Ans.4 "Should" indicates a recommendation that is not required.

American Society of Safety Engineers (2009). Criteria for Acceptable Practices in Safety, Health and Environmental Training, ANSI/ASSE Z490.1, (1st ed.). American National Standards Institute. 10.

2140. In ANSI Z490.1, Criteria for Accepted Practices in Safety, Health, and Environmental Training, when 'shall' is used in the standard it indicates a statement that is:

1) permissive.
2) mandatory.
3) a requirement that is strongly suggested.
4) a recommendation.

Ans.2 "Shall" indicates a mandatory requirement.

American Society of Safety Engineers (2009). Criteria for Acceptable Practices in Safety, Health and Environmental Training, ANSI/ASSE Z490.1, (1st ed.). American National Standards Institute. 11.

2141. Under the ANSI Z490.1 standard, the Annexes of the standard are provided for:

1) further clarification of the meaning of the standard and clear direction on how to comply with the standard.
2) useful information that may be of use to particular audiences with varying levels of usefulness.
3) use by all persons and organizations that will implement the standard in developing their training programs.
4) implementation by the Committee members when determined appropriate.

Ans.2 Under the ANSI Z490.1 standard, the Annexes of the standard are provided for useful information that may be of use to particular audiences with varying levels of usefulness.

American Society of Safety Engineers (2009). Criteria for Acceptable Practices in Safety, Health and Environmental Training, ANSI/ASSE Z490.1, (1st ed.). American National Standards Institute. 29.

2142. The ANSI/ASSE Z490.1 - 2009, Criteria for Acceptable Practices in Safety, Health and Environmental Training uses a two column format. In this format:

1) The left column establishes the requirements of the standard and the right column provides explanatory information.
2) The left column provides the requirement headings and the right column explains the specific actions necessary to meet the standard.
3) The left column establishes the requirements approved by the committee and the right column provides the opinions of the dissenting members.
4) There is no two column format in the ANSI standards system.

Ans.1 The ANSI/ASSE Z490.1 - 2009, Criteria for Acceptable Practices in Safety, Health and Environmental Training uses a two column format. The left column establishes the requirements of the standard and the right column provides explanatory information.

American Society of Safety Engineers (2009). Criteria for Acceptable Practices in Safety, Health and Environmental Training, ANSI/ASSE Z490.1, (1st ed.). American National Standards Institute. 7.

2143. Training that lacks in _____ will result in slower improvement, confusion or deviation from expectations.

1) Consistency
2) Facts
3) Direct Applicability
4) Trust
5)

Ans.1 Consistency is critical to training to provide for clear understanding and maximizing performance in line with expectations.

Baldwin, D.A. (1992). Safety and Environmental Training: Using Compliance to Improve Your Company, (1st ed.). Van Nostrand Reinhold. 16.

2144. An objective for the course states that, "Students will be able to apply systems engineering concepts to solve the challenges of project management. " Which objective types does this objective best illustrate?

1) Behavior based objectives
2) Content based objectives
3) Conditionally based objectives
4) Problem-centered objectives

Ans4. Problem-centered objectives address problem solving skills requiring the application of complex intellectual skills often involving the use of pre-existing knowledge and skills.

Galbraith, M.W. (2004). Adult Learning Methods: A Guide for Effective Instruction, (3rd ed.). Krieger Publishing Company. 106.

2145. Which of the following is an activity where individuals generate ideas related to a topic or task; done in groups or individually with no restriction on quality of ideas?

1) Roundtable
2) Discussion Group
3) Brainstorming
4) Isolationism action

Ans.3 Brainstorming is an activity where individuals generate ideas related to a topic or task; done in groups or individually with no restriction on quality of ideas. Once ideas are generated, they are then evaluated and a decision about which to pursue is made.

2146. Which of the following is an outcome statement?

1) Develop safe behavior.
2) Provide instruction to the proper management of hazardous waste.
3) Conduct practical exercises in spill response.
4) Be able to properly don a respirator.

Ans.4 The correct answer describes an outcome, something that the student is able to achieve and continue to achieve after the instructional process is complete. The other three options describe the process of teaching, training or instruction.

Mager, R.F. (1997). Preparing Instructional Objectives, (3rd ed.). CEP Press. 23.

2147. Which step of the training cycle is testing the training participants prior to the beginning of class?

1) Evaluation
2) Analysis
3) Design and Development
4) Implementation

Ans.4 Analysis includes the validation of training needs and objectives, the assessment of needs, or even if training is needed.

Design and development incorporates the identifying and assembling the methodology and materials needed to accomplish the training. Design should be based on the findings of the analysis.

Implementation involves the actual performance of the training activity whether it is classroom instruction, self-paced, on-the job or other form of training.

Evaluation is any process that determines if the objectives were accomplished and the desired outcomes were achieved. Evaluation occurs after (and sometimes during) training and can feed into the beginning of the next training cycle, testing participants at the end of training.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 42.

2148. Which step of the training cycle is testing participants at the end of training?

1) Evaluation
2) Analysis
3) Design and Development
4) Implementation

Ans.1 Analysis includes the validation of training needs and objectives, the assessment of needs, or even if training is needed.

Design and development incorporates the identifying and assembling the methodology and materials needed to accomplish the training. Design should be based on the findings of the analysis.

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Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 43.

2149. Of which of the learning categories does a video and practice fall in to?

1) Participatory
2) Knowledge
3) Skills
4) Corrective

Ans.3 The three learning categories are knowledge, skills and attitude. Examples of ways to gain knowledge are: audiotapes, lectures, articles, and diagrams. Examples of ways to impart new skills are: demonstrations, role playing, videos and practice, exercises and worksheets. Examples of tools to change values are: debates, structured games, self-analysis and exercises.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 30.

2150. The classic college final examination is an example of which level of learning evaluation?

1) Results
2) Learning
3) Reaction
4) Behavior

Ans.2 Learning - measure the extent to which learning has occurred.

Don Kirkpatrick's four levels of training evaluation are reaction, learning, behavior, and results.

Reaction - measures participant satisfaction with the training.

Learning - measure the extent to which learning has occurred.

Behavior - measures whether the skills and knowledge learned are being implemented.

Results - measures the business impact of the implemented skills and knowledge.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 254.

2151. What type of evaluation method is typically accomplished through the use of true/false questions, matching lists or fill-in-the-blank questions?

1) Course completion evaluation
2) Knowledge, skills and abilities assessment
3) Objective tests
4) Needs/objectives/content comparison

Ans.3 The following is a selection of evaluation methods:

Objective tests - specific questions directed at measuring the trainees learning of the information specified in the course objectives. Usually quantitative in nature.

Attitude surveys - used to identify changes in attitudes usually related to work habits, motivation, values, beliefs and working relationships. Qualitative in nature.

Simulation and observation - an instructor observes the trainee in a job simulation to determine if the desired skills or behaviors have been learned.

Productivity or performance reports - real time production data is used to assess performance improvement after training.

Post-training surveys - are used to assess the retention and implementation of knowledge or skills learned at training. Subjective and qualitative in nature.

Needs/objectives/content comparison - used to determine if a program was relevant to the needs. This is effectively a cross walk of the needs to the objectives and the objectives to that actual training content.

Evaluation forms - used to collect participant input on their 'satisfaction' with a program.

Interviews - useful for assessing skills and knowledge understanding and implementation when observation is not viable.

Instructor evaluation - used to measure an instructors effectiveness, competence and instructional skills.

Biech, E. (2005). Training for Dummies, (1st ed.). Wiley Publishing, Inc.. 263.

2152. When broken down to the basics, learning is:

1) nothing more than the transfer of information from the source to the learner.
2) an activity that takes place in a classroom type environment.
3) a transformation where the leaner's mind is different.
4) a complex process.

Ans.3 The best answer is that learning is a transformation where the leaner's mind is different. The purpose of training is to bring about change.

Stolovitch, H.D., & Keeps, E.J. (2011). Telling Ain't Training, (2nd ed.). ASTD Press. 13.

2153. With regard to learner evaluation, which of the following is most accurate?

1) To be effective, learner evaluations should be accomplished as soon as possible after the training has been completed. This allows the students to reinforce what they have learned in class.
2) Effective learner evaluation can occur before, during and immediately following the training or even some time after the training has been completed.
3) Effective post training evaluation has to be supported by an appropriate pre-training exam, thus allowing a true measure of what the learner has accomplished through the course.
4) Mid-training evaluation is limited in effectiveness and usefulness. Typically, mid-training evaluation is performed to provide a transition from one subject to another.

Ans.2 Learner evaluations can be performed before, during and after the training activity. Post training evaluation may be accomplished immediately after the training or sometime later. The time period in which evaluations are accomplished depends on the purpose of the evaluation.

Galbraith, M.W. (2004). Adult Learning Methods: A Guide for Effective Instruction, (3rd ed.). Krieger Publishing Company. 111.